VOILA: Value-of-Information Guided Visual Fidelity Selection for Bandwidth-Constrained settings

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***Abstract*—Visual evidence is commonly acquired at a fixed fidelity even when the query does not require all of its detail. This design pays the transmission or retrieval cost before a remote vision–language model (VLM) can decide whether that detail was useful. We study *pre-retrieval fidelity selection*: given the question and cached metadata, choose which visual representation to acquire before accessing its pixels or invoking the VLM. The central hypothesis is that linguistic structure predicts visual information need. VOILA learns a success predictor for each fidelity, calibrates the predictors independently, and selects the representation maximizing predicted correctness minus its measured byte cost. We evaluate six VLMs on the same 1445-query FloodNet population using five image-grouped folds with disjoint training, calibration, operating-point validation, and test images. VOILA obtains 70.5% mean accuracy at 0.136 MB/query, reducing bytes by 97.5% relative to transmitting full images. Against the best fixed fidelity, it matches one-second deadline-qualified accuracy while reducing mean payload by roughly 62%, with a 0.6–1.7 point accuracy cost on four models and accuracy gains on LLaVA and Pixtral. The benefit is not universal: a strong fixed thumbnail policy is best for Qwen2-VL-72B, and unseen question templates reduce accuracy by 17.4–37.1 points. These results show when language can guide information acquisition before transmission, and expose calibration, model preprocessing, and linguistic shift as the principal constraints.**

I. Introduction

Multimodal systems increasingly separate where visual data is captured from where it is interpreted. Images may be retained as captions, thumbnails, compressed previews, and full-resolution originals, while an expensive VLM is invoked remotely [1]–[4]. The usual pipeline acquires the highest-fidelity representation first and asks whether it was necessary afterward. This order hides an important decision: *what evidence should be retrieved or transmitted before inference?*

Flood response provides a concrete instance. Aerial capture can produce images faster than a disrupted or congested uplink can deliver them, while analysts ask questions such as *"What is the overall condition of this area?"* and *"How many buildings are flooded?"* [5]. The former often depends on coarse scene semantics; the latter depends on small-object detail. Sending every original preserves the possibility of answering either question, but consumes the scarce link before the question's information need is considered. Sending every thumbnail saves bytes but can destroy the evidence required by counting and localization.

This motivates a pre-retrieval, or equivalently pre-transmission, stage that is orthogonal to model routing. Model cascades and early-exit methods reduce the computation applied after context is available [6]–[10]. Compression, adaptive video analytics, edge offloading, semantic communication, and visual-token reduction reduce representation or compute after inspecting data or executing part of the pipeline [11]–[19]. Neither answers which cached representation should cross the storage or network boundary in the first place.

The core question is therefore: **can the language of a question predict the visual fidelity that a remote VLM will need?** Two questions over the same asset can express very different evidence requirements. Phrases such as *how many* imply enumeration of small instances, whereas *overall condition* asks for scene-level recognition. Prior work shows that question form predicts VQA difficulty [20], [21] and that query features can predict retrieval quality before retrieval [22]–[24]. We investigate whether this signal is strong enough to select visual fidelity under an actual byte budget.

The decision is harder than ordinary classification. Correctness is non-monotone in fidelity for modern VLMs; a compressed representation can outperform an original by suppressing irrelevant detail. Raw success scores cannot be compared as probabilities across fidelities, and VLM log probabilities are available only after paying the acquisition and inference cost. A useful policy must estimate counterfactual success before image access, calibrate those estimates on disjoint data, and attach them to the measured size of each cached representation.

We introduce VOILA, a language-conditioned, value-of-information policy for this stage. For every candidate fidelity, a lightweight predictor estimates whether the target VLM would answer correctly. An independent isotonic map turns each score into a probability, and a validation-selected byte penalty chooses the representation with the largest predicted correctness minus exact payload cost. The selected representation is acquired once and the remote VLM is invoked once.

This paper makes four contributions:

- We formulate pre-retrieval visual fidelity selection as cost-sensitive information acquisition, distinct from post-retrieval model routing.
- We show on FloodNet that question form predicts fidelity need, while no fixed fidelity is uniformly best across six VLMs.
- We develop a leakage-resistant calibrated selection pipeline and a validation-only operating-point rule using measured, item-specific bytes.
- We evaluate accuracy–byte frontiers, matched-policy comparisons, mission budgets, and deadline-qualified accuracy using six public cellular scenarios, while reporting the cases in which fixed or supervised routing remains stronger.

TABLE I
FIXED-FIDELITY ACCURACY (%) UNDERLYING FIGURE 1. THE BEST INPUT IS MODEL-DEPENDENT AND IS NEVER THE FULL IMAGE IN THIS POPULATION.

Model	Caption	32px	Q1	Q10	Full
Llama-4-Scout	54.2	24.6	64.7	70.2	63.1
LLaVA-7B	73.3	73.8	42.9	75.6	74.7
Pixtral-12B	17.7	20.2	28.0	50.5	38.8
Qwen-235B	66.0	25.8	75.9	83.2	76.6
Qwen2-VL-72B	30.7	73.9	23.6	39.8	58.8
Qwen2-VL-7B	42.8	59.7	64.5	71.5	67.4

II. MOTIVATING OBSERVATION

Question form separates visual information requirements. Figure 1 (left) groups FloodNet questions using only their surface form. Across all six VLMs, questions beginning with *how many* gain as more visual detail becomes available. Condition-recognition questions instead peak at JPEG Q10 on average and frequently succeed from a cached caption or thumbnail. This is not a claim that language determines the answer; it predicts how much visual evidence is likely to be useful before that evidence is acquired.

More pixels do not imply a better answer. Figure 1 (right) shows the five fixed-fidelity curves. Their non-monotonicity is substantial: LLaVA is strong on captions and 32-pixel images, Qwen2-VL-72B peaks at 32 pixels, and Pixtral benefits from JPEG Q10. Full resolution is not the most accurate fixed input for any evaluated model. The selection problem is therefore not reducible to estimating the smallest image that approximates the original; it must estimate VLM success directly.

Post-hoc confidence arrives too late and can be misleading. A cascade must first retrieve a representation, run the VLM, and then inspect its confidence. When visual grounding is weak, fluent answers driven by language priors can still receive high probability. This is precisely the failure mode observed for the cumulative-cost Log-prob Cascade in Section VI: confidence-based escalation is both post-retrieval and poorly aligned with missing visual information. These observations motivate a language-only predictor whose scores are explicitly calibrated for comparing fidelities.

III. PROBLEM FORMULATION

Let q be a question associated with a visual asset x, and let $\mathcal{F} = \{f_1, \ldots, f_K\}$ be cached representations ordered by nominal fidelity. In our implementation these are an offline cached caption generated once at ingestion, a 32×32 PNG, JPEG Q1, JPEG Q10, and the original image. The client can read each representation's stored byte count $b_f(x)$ from metadata without accessing its content. Let M be a fixed remote VLM and $Y_f(q, x) \in \{0, 1\}$ indicate whether M answers correctly when given f.

The pre-retrieval policy does not observe any Y_f at decision time. It must estimate

$$p_f(q) = \Pr[Y_f(q, x) = 1 \mid \phi(q)], \tag{1}$$

where $\phi(q)$ contains only linguistic features. Given a byte price $\lambda \geq 0$, the one-step value-of-information decision is

$$f^*(q, x) = \arg\max_{f \in \mathcal{F}} [\widehat{p}_f(q) - \lambda b_f(x)]. \tag{2}$$

This objective treats an expected correct answer as unit utility and charges the exact bytes that cross the boundary. Other deployments may substitute a monetary, energy, or latency cost without changing the estimator.

Three properties make Eq. 2 difficult. First, its labels are counterfactual: offline training must evaluate every fidelity, but online selection is useful only if it avoids doing so. Second, a raw classifier score for captions is not commensurate with a raw score for full images. Utility subtraction is meaningful only when each is mapped to a probability on held-out data [25], [26]. Third, images and repeated question templates create leakage paths. Random row splits can place nearly identical evidence on both sides of evaluation. We therefore separate images across all stages and test template shift independently.

Calibration matters because errors are paid twice: an underestimated useful fidelity can remove answer evidence, while an overestimated one wastes bytes. The following standard decision-regret argument makes the dependence explicit.

Proposition 1 (Selection regret under calibrated estimation)**.** *Let p_f be the true conditional success probabilities and suppose $|\widehat{p}_f - p_f| \leq \epsilon$ for every $f \in \mathcal{F}$. If f^* maximizes the true utility $p_f - \lambda b_f$ and $\widehat{f}$ maximizes $\widehat{p}_f - \lambda b_f$, then*

$$(p_{f^*} - \lambda b_{f^*}) - (p_{\widehat{f}} - \lambda b_{\widehat{f}}) \leq 2\epsilon.$$

Proof. Add and subtract the two estimated utilities. The middle difference is non-positive because $\widehat{f}$ maximizes estimated utility; each remaining probability error is at most ϵ. □

The result does not assert that isotonic regression always has the smallest finite-sample error. It states why the system needs probabilities whose errors are controlled on data separate from both predictor fitting and operating-point selection. The calibration ablation in Section VI tests that choice rather than assuming it.

IV. SYSTEM DESIGN

A. Offline success learning

For a target VLM, the offline stage records correctness at every fidelity on labeled examples. We extract sparse TF–IDF features and structural indicators for counting, condition, yes/no form, numerals, negation, and question length. A fidelity-specific gradient-boosted predictor g_f produces a score $r_f(q) = g_f(\phi(q))$. The predictors model the behavior of the target VLM; they do not replace it and never consume the visual representation.

The feature representation deliberately separates lexical and structural cues: TF–IDF captures distinctions between repeated question forms, while structural features represent properties expected to transfer beyond a single vocabulary. Examples include an initial *how many*, a yes/no auxiliary, numeric tokens, negation, and question length. This design tests the linguistic

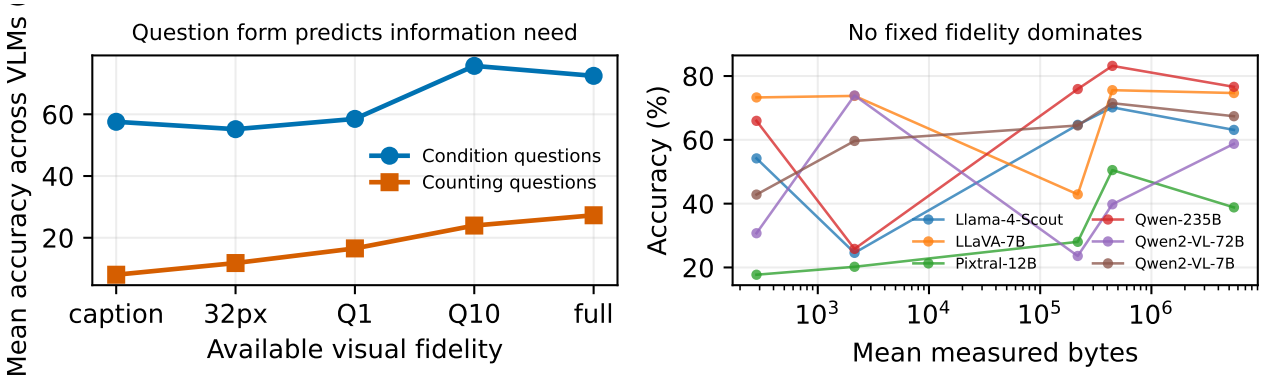


Fig. 1. The motivating observation on the common FloodNet population. Left: mean VLM accuracy separated only by the linguistic form of the question; counting benefits most from detail, while condition recognition peaks at a compressed fidelity. Right: fixed-fidelity curves differ sharply across the six VLMs, so neither full resolution nor one compressed tier is a universal choice. Values are computed from cached answers on the same image-grouped population used in the main evaluation.

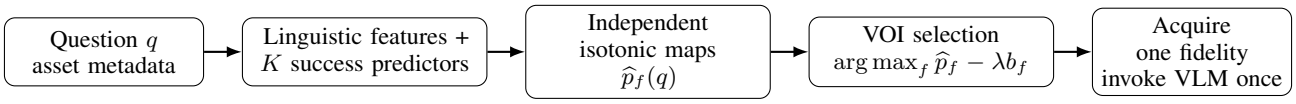


Fig. 2. VOILA deployment topology. An edge client or gateway holds a compact metadata index over cached cloud/archive representations and chooses one payload before transmission to the remote VLM. The decision reads the question and byte metadata, but no pixels. Caption generation occurs once during ingestion; online accounting includes only the selected cached payload.

hypothesis directly and keeps the selection stage independent of a second large language model. A richer encoder remains a baseline rather than an unpriced component of VOILA.

We fit an independent isotonic calibrator ψ_f on a population disjoint from predictor training:

$$\widehat{p}_f(q) = \psi_f(r_f(q)). \tag{3}$$

Independent maps are necessary because score scales and error rates differ by fidelity. Isotonic regression preserves each predictor's rank while learning a non-parametric mapping to empirical correctness [25]. It does not impose cross-fidelity monotonicity, because Figure 1 shows that VLM correctness is not monotone in pixels.

B. Validation-only operating point

For each outer fold, a separate validation population selects λ. Let $A^{\text{val}}_{\text{fix}}$ be the accuracy of the best fixed fidelity on validation data. Among candidate penalties whose validation accuracy is at least $0.95A^{\text{val}}_{\text{fix}}$, we choose the one with the smallest measured mean bytes. Neither the test answers nor test costs choose the operating point. This rule yields a deployable point rather than selecting the most attractive location after seeing the test frontier.

C. Online selection and system boundary

Algorithm 1 summarizes inference. The client reads q and the already-indexed byte lengths, evaluates the lightweight predictors and calibrators, and acquires exactly one representation. The remote VLM is then called once. The boundary excludes one-time caption generation, because captions are cached during ingestion, and includes the exact selected payload. The

Algorithm 1 VOILA pre-retrieval fidelity selection

Require: question q, asset metadata $\{b_f(x)\}_{f\in\mathcal{F}}$, predictors g_f, calibrators ψ_f, validation-selected λ

for all $f \in \mathcal{F}$ **do**
  $r_f \leftarrow g_f(\phi(q))$
  $\widehat{p}_f \leftarrow \psi_f(r_f)$
  $u_f \leftarrow \widehat{p}_f - \lambda b_f(x)$
end for
$f^* \leftarrow \arg\max_{f\in\mathcal{F}} u_f$
acquire x at f^*; invoke the remote VLM once
return the VLM answer and selected fidelity f^*

design composes with model routing: after acquisition, another policy may still choose which remote model processes the evidence.

The intended topology is an edge–cloud or archive–cloud split rather than an on-drone VLM deployment. A drone, phone, or field gateway has the user's question and an index containing asset identifiers, byte counts, and cached caption pointers. The high-volume image tiers reside in remote object storage or at a nearby gateway. VOILA runs at the requester or gateway, selects one tier, and only then transmits that payload to the VLM service. This is why Figure 2 is a placement diagram as well as a pipeline: the decision is made before the expensive network/storage boundary, not between VLM layers.

The algorithm differs from a sequential cascade in both action and accounting. It evaluates all K lightweight success predictors, but it does not acquire K visual inputs and does not invoke the VLM K times. It also allows the selected fidelity to skip intermediate tiers. This is essential when a caption

TABLE II
MEASURED CACHED REPRESENTATION SIZES ON THE COMMON FLOODNET POPULATION. THE ROUTER READS THESE BYTE COUNTS AS METADATA.

Representation	Median	p95	Mean bytes
Cached caption	188	663	281
32$imes$32 PNG	2,125	2,605	2,141
JPEG Q1	213,991	248,887	217,576
JPEG Q10	432,120	638,742	445,479
Full image	5,195,608	7,622,159	5,520,774

and an original are cached independently: the cost of choosing the original is its payload, not the sum of every lower-quality representation. We retain the cumulative rule only as a legacy ablation.

V. EXPERIMENTAL METHOD

The evaluation answers three questions. **RQ1** asks whether linguistic pre-retrieval selection improves the measured accuracy–byte frontier over fixed and adaptive policies. **RQ2** asks whether byte reductions produce more correct answers under upload deadlines and shared mission budgets. **RQ3** asks which calibration and feature choices enable the result, and where the linguistic assumption fails.

A. Driving use case and population

We use FloodNet as the driving case because it pairs aerial flood imagery with questions whose visual requirements range from scene condition to object counting [5]. The common population contains 1445 query records for which all five cached fidelities and all six VLM outputs are available. The models span LLaVA-7B, Pixtral-12B, Llama-4-Scout, Qwen2-VL-7B, Qwen2-VL-72B, and Qwen-235B [4], [27]–[30]. We make no new VLM or API calls; cached outputs are the only source of answer correctness.

Table II reports measured bytes rather than a normalized proxy. Each outer test fold holds out images, not rows. The remaining images are split again into mutually disjoint predictor-training, calibration, and operating-point-validation populations. We use five outer folds and report query-level selections for every policy. Confidence intervals and paired differences resample image clusters, keeping questions from the same image together.

This separation is stricter than a conventional train/test split. Predictor training learns g_f; calibration alone fits ψ_f; validation alone chooses λ and the best fixed comparator; and the test fold is used once for reporting. Because the unit of assignment is the image, questions sharing the same visual evidence cannot cross these boundaries. The leave-template-out stress test then changes the grouping variable to ask a different question: whether the learned linguistic regularities survive unseen phrasing.

B. Policies and metrics

We compare all five fixed fidelities; the cross-validated best fixed fidelity; a cumulative-cost Log-prob Cascade inspired by FrugalGPT [6]; a supervised BERT Router; the earlier sequential VOILA rule as an ablation; and a cached hindsight oracle. The GPT-4o router is omitted because its cached labels do not cover the complete common population. The primary metrics are correctness, exact transmitted bytes, bytes at matched accuracy, and accuracy at matched cost.

For network consequences, we simulate one-way evidence serialization using six public CellReplay scenarios [31]: T-Mobile driving, stationary-good, and weak-signal; and Verizon crowded, stationary-good, and walking. The simulator consumes the recorded `up-heavy-pdo`, `up-delay-light-pdo`, and `latency-offset-up` streams as 1,400-byte delivery opportunities. For each query it samples 100 deterministic, uniform trace phases with seed 2027 and wraps each trace circularly. It adds no synthetic loss. We report transfer p50/p95, completion by 1, 5, and 10 seconds, and deadline-qualified accuracy

$$\mathrm{DQA}(D) = \Pr[Y = 1 \ \wedge \ T_{\mathrm{upload}} \leq D]. \tag{4}$$

This is a recorded-trace uplink serialization simulation, not TCP, end-to-end CellReplay, VLM latency, or operator response time.

The mission experiment allocates a shared byte budget over deterministic batches for Qwen-235B. The CPU benchmark measures the saved router on one reference CPU after warm-up. We also hold out complete question templates to test whether lexical regularity, rather than transferable semantics, drives the policy.

VI. RESULTS

A. Does language improve the accuracy–byte frontier?

Language-conditioned selection occupies a useful but not universal frontier. Figure 3 and Table III show that VOILA averages 70.5% at 0.136 MB/query, compared with 63.2% at 5.521 MB for full transmission. It improves on full-image accuracy by 7.3 points on average because full resolution is not the VLM accuracy oracle. The advantage is model-dependent: VOILA is above best fixed for Pixtral, but best fixed is 0.6–1.7 points higher for four models. The claim is therefore byte efficiency at a validated accuracy floor, not universal accuracy dominance.

Measured bytes expose wins hidden by nominal fidelity labels. Table III connects each accuracy to the payload actually sent. Qwen2-VL-72B reaches 73.3% at roughly 1.9 KB/query; LLaVA reaches 75.7% at 0.015 MB/query; Pixtral requires 0.242 MB/query for 54.3%. A policy described only as choosing "low," "medium," or "high" fidelity would miss this two-order-of-magnitude variation across representations and assets.

Matched comparisons distinguish savings from accuracy trade-offs. Table IV uses paired image-clustered resampling. VOILA is above the best fixed point for LLaVA and Pixtral and below it by 0.6–1.7 points for four models, while using fewer bytes in every paired comparison. At matched or higher accuracy it saves up to 2.37 MB/query when such a fixed point exists. The Qwen2-VL-72B case is a cautionary example for matched-cost reporting: VOILA averages 1,916 bytes/query

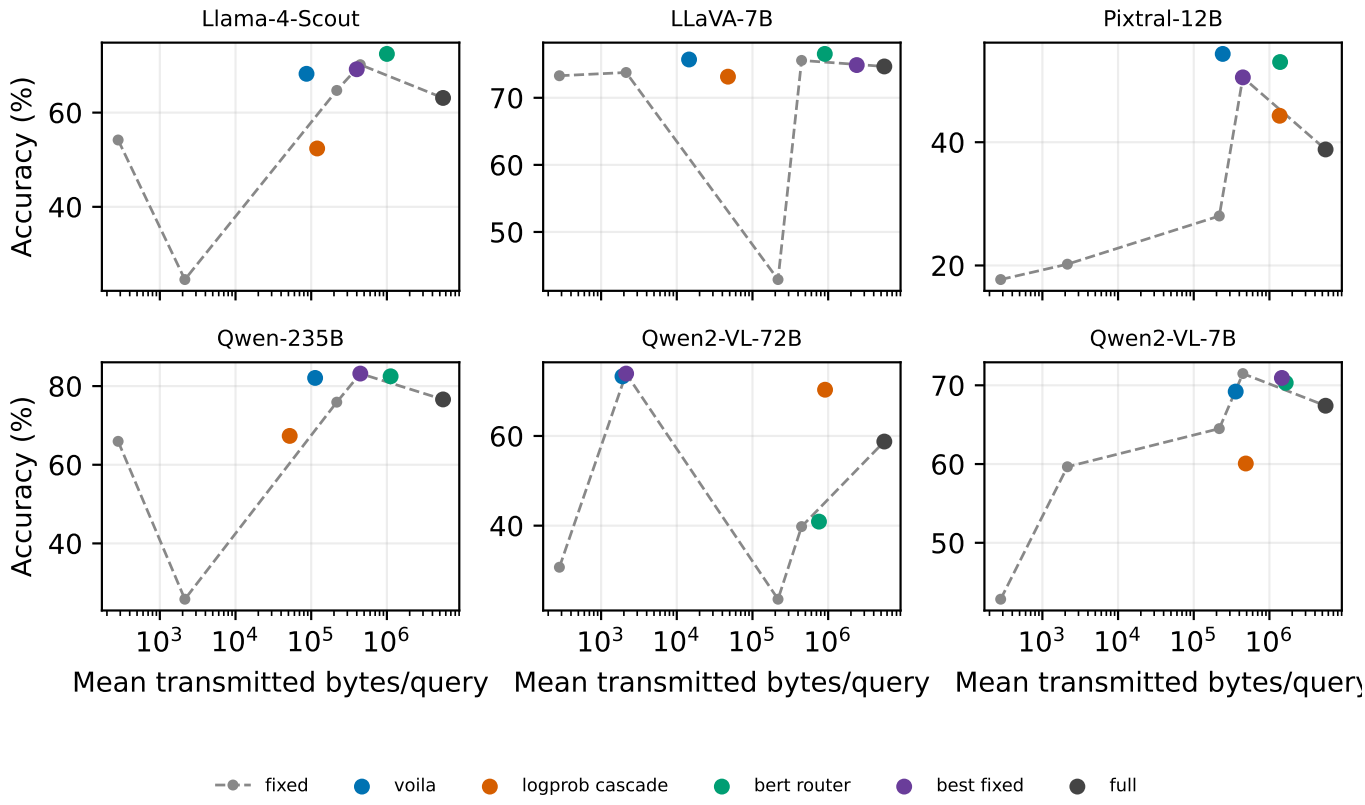


Fig. 3. Measured-byte FloodNet frontier. Every point uses the same image-grouped test population. Fixed points trace the five cached fidelities; the cascade includes every representation acquired before stopping.

TABLE III
FLOODNET ACCURACY (%) AND MEASURED MB/QUERY. BEST FIXED IS SELECTED WITHOUT THE HELD-OUT FOLD; CASCADE COST IS CUMULATIVE.

Model	Full acc.	MB	Best fixed acc.	MB	Cascade acc.	MB	BERT acc.	MB	VOILA acc.	MB
Llama-4-Scout	63.1	5.521	69.2	0.400	52.4	0.120	72.5	1.000	68.2	0.087
LLaVA-7B	74.7	5.521	74.9	2.386	73.1	0.047	76.5	0.903	75.7	0.015
Pixtral-12B	38.8	5.521	50.5	0.445	44.3	1.364	53.0	1.384	54.3	0.242
Qwen-235B	76.6	5.521	83.2	0.445	67.3	0.052	82.5	1.120	82.1	0.113
Qwen2-VL-72B	58.8	5.521	73.9	0.002	70.3	0.910	40.9	0.756	73.3	0.002
Qwen2-VL-7B	67.4	5.521	70.9	1.461	60.1	0.487	70.3	1.641	69.2	0.358

TABLE IV
PAIRED VOILA DIFFERENCES AGAINST BEST FIXED (BF) AND BERT. POSITIVE ΔACCURACY FAVORS VOILA; NEGATIVE ΔMB MEANS FEWER BYTES.

Model	Δacc BF	ΔMB BF	Δacc BERT	ΔMB BERT
Llama-4-Scout	-1.0	-0.313	-4.2	-0.913
LLaVA-7B	+0.8	-2.372	-0.8	-0.888
Pixtral-12B	+3.8	-0.204	+1.3	-1.142
Qwen-235B	-1.1	-0.333	-0.4	-1.008
Qwen2-VL-72B	-0.6	-0.000	+32.4	-0.754
Qwen2-VL-7B	-1.7	-1.103	-1.1	-1.282

TABLE V
IMAGE-CLUSTERED 95% BOOTSTRAP INTERVALS FOR THE SELECTED VOILA OPERATING POINT. BYTES ARE MB/QUERY.

Model	Accuracy	95% CI	MB	95% CI
Llama-4-Scout	68.2	[65.9,70.7]	0.087	[0.077,0.096]
LLaVA-7B	75.7	[73.4,78.0]	0.015	[0.011,0.018]
Pixtral-12B	54.3	[51.8,57.0]	0.242	[0.230,0.254]
Qwen-235B	82.1	[80.2,83.9]	0.113	[0.103,0.122]
Qwen2-VL-72B	73.3	[71.0,75.5]	0.002	[0.002,0.002]
Qwen2-VL-7B	69.2	[66.8,71.6]	0.358	[0.336,0.383]

and the fixed 32-pixel tier averages 2,141 bytes/query, producing 73.3% versus 73.9% accuracy at effectively the same transmission cost. Treating only strictly lower-cost fixed tiers as "affordable" would compare against the 281-byte caption tier and create a spurious 42.6-point gain from a 225-byte gap. BERT is more accurate on Llama-4 and slightly more accurate on several Qwen/LLaVA settings, but uses about 0.90–1.64

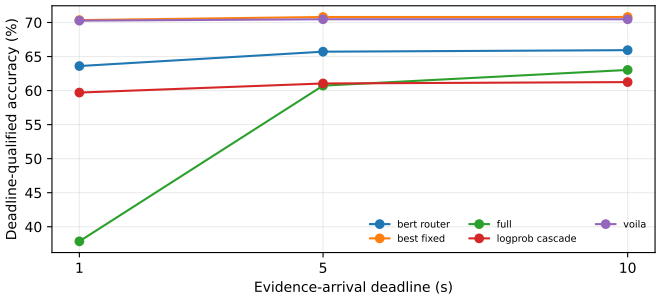


Fig. 4. Deadline-qualified accuracy averaged over six recorded cellular scenarios and six VLMs. A response counts only when it is correct and its evidence has arrived by the deadline.

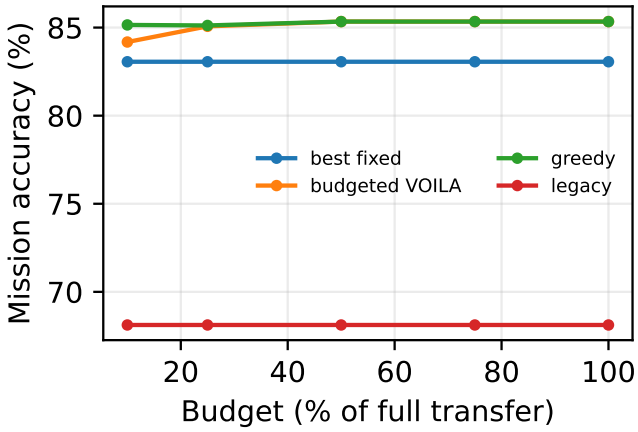


Fig. 5. Qwen-235B mission accuracy for 100-query batches under a shared byte budget. Budget is expressed as a fraction of sending every original.

MB/query on those models. Pixtral is the clearest adaptive win: 54.3% at 0.242 MB for VOILA versus 50.5% at 0.445 MB for best fixed.

VLM confidence is a weak substitute for pre-retrieval prediction. The orange points in Figure 3 show that the Logprob Cascade is cheaper than VOILA for some models but sacrifices substantial accuracy, including 67.3% versus 82.1% on Qwen-235B. On Pixtral it sends 1.364 MB/query yet reaches only 44.3%, below VOILA at both axes. Because cascade bytes include every attempted fidelity, post-hoc escalation cannot recover the acquisition cost that pre-retrieval selection avoids.

The effect is stable at the image level even when model gaps are small. Table V reports intervals obtained by resampling images rather than individual questions. The narrow byte intervals show that the selected payload mix is not determined by a few unusually large originals. Accuracy uncertainty is wider for the smaller per-model test folds, which is why Table IV reports paired differences instead of ranking methods from point estimates alone.

B. Do byte savings improve timely and mission-level utility?

Byte savings become deadline-qualified answers under short deadlines. Figure 4 and Table VI show that VOILA reaches 70.3% DQA at one second, while full-image transmission reaches 37.9%. Its mean p50/p95 serialization latencies are 0.052/0.230 seconds, compared with 1.428/2.959 seconds for full images. At five seconds the gap narrows because most full payloads arrive; this is expected, since routing changes transmission time rather than offline answer correctness.

A strong fixed policy remains competitive when its payload is already small. Table VI shows best fixed at 70.35% one-second DQA, about 0.08 point above VOILA, and 70.81% at ten seconds, about 0.34 point above. The adaptive benefit appears primarily in bytes and tail latency: VOILA reduces mean payload from roughly 0.36–0.37 MB for the compact baselines to 0.136 MB and reduces p95 serialization to 0.23 seconds. Language conditioning matters most when the query mix or available fixed point does not already align with the link.

Thus the headline 97.5% byte reduction and 37.9% to 70.3% one-second DQA improvement are relative to full-image transmission. The comparison to best fixed is smaller and more operationally specific: essentially identical one-second DQA, lower tail latency, and about a 62% mean-payload reduction.

The deadline result is not driven by one favorable trace. Table VII separates all six carrier/scenario pairs. Full-image DQA varies sharply with the recorded opportunity process, whereas VOILA remains close to its offline correctness ceiling and keeps p95 serialization below one second in every aggregated scenario. The table should still be read as a sensitivity analysis over public cellular traces, not as a carrier or disaster-network comparison.

Batch allocation favors explicit marginal-value sorting. Figure 5 and Table VIII show that budgeted VOILA reaches 84.2% accuracy at 10% of full-transfer bytes, 1.1 points above best feasible fixed but 1.0 point below value-per-byte greedy. At 25% the adaptive optimizers are effectively tied at 85.1%, and at 50% both reach 85.3%. We therefore do not present the calibrated per-query policy as the best batch allocator. When a whole batch and a hard shared budget are known in advance, sorting predicted marginal gains per byte is preferable; the per-query VOI rule is the online policy for independent arrivals or unknown future demand. The cached oracle reaches 90.5% at the 10% budget, leaving meaningful headroom in predicted marginal value.

C. What enables the gain, and where does it fail?

Calibration makes the utility comparable, but isotonic is not uniformly the most accurate variant. Tables X and IX report the outcomes and the calibrator recommended on held-out validation data. Independent isotonic is the primary definition because Eq. 2 requires per-fidelity probabilities and it avoids sharing one scale across heterogeneous representations. In the ablation, however, no calibration is slightly better for Pixtral, Qwen-235B, Qwen-72B, and Qwen-7B in the selected operating point, while validation recommends shared Platt for LLaVA and independent isotonic for Llama-4 and Qwen-72B. The theory motivates calibrated probabilities for

TABLE VI
RECORDED-TRACE ONE-WAY UPLINK SERIALIZATION RESULTS AVERAGED ACROSS MODELS AND SCENARIOS. LATENCIES ARE SECONDS; C@1 IS ONE-SECOND COMPLETION AND DQA IS DEADLINE-QUALIFIED ACCURACY.

Policy	MB	p50	p95	C@1	DQA@1	DQA@5	DQA@10
Full	5.521	1.428	2.959	59.3	37.9	60.7	63.0
Best fixed	0.359	0.112	0.281	99.5	70.3	70.8	70.8
Cascade	0.497	0.034	1.237	96.3	59.7	61.0	61.2
BERT	1.134	0.054	2.370	91.7	63.6	65.7	65.9
VOILA	0.136	0.052	0.230	99.7	70.3	70.5	70.5

TABLE VII
ONE-SECOND DQA BY RECORDED SCENARIO. BF IS BEST FIXED; LATENCY IS VOILA P95 SERIALIZATION TIME IN SECONDS.

Carrier	Scenario	Full DQA	BF DQA	VOILA DQA	VOILA p95
Tmobile	driving	32.3	70.1	69.9	0.308
Tmobile	stationary-good	54.8	70.8	70.5	0.092
Tmobile	weak-signal	0.0	68.7	69.8	0.674
Verizon	crowded	22.8	70.8	70.5	0.111
Verizon	stationary-good	62.8	70.8	70.5	0.079
Verizon	walking	54.4	70.8	70.5	0.114

TABLE VIII
QWEN-235B MISSION ACCURACY (%) FOR BATCHES OF 100.

Policy	10%	25%	50%
Best fixed	83.1	83.1	83.1
Budgeted VOILA	84.2	85.1	85.3
Greedy	85.2	85.1	85.3
Legacy	68.1	68.1	68.1

TABLE IX
CALIBRATION, FEATURE, AND TEMPLATE-HOLDOUT STRESS RESULTS. MAIN AND TEMPLATE COLUMNS ARE ACCURACY (%); "STRUCT ONLY" REMOVES TF–IDF. THE FINAL COLUMN IS THE VALIDATION-RECOMMENDED CALIBRATOR.

Model	Main	Template	Δ	Struct only	Cal.
Llama-4-Scout	68.2	45.7	-22.6	68.4	iso
LLaVA-7B	75.7	58.3	-17.4	76.1	shared_platt
Pixtral-12B	54.3	26.2	-28.1	53.8	none
Qwen-235B	82.1	45.0	-37.1	81.1	none
Qwen2-VL-72B	73.3	44.2	-29.1	73.4	iso
Qwen2-VL-7B	69.2	48.8	-20.4	68.7	none

TABLE X
CALIBRATION ABLATION AS ACCURACY (%)/MB PER QUERY. EACH VARIANT USES THE SAME SPLIT AND VALIDATION-ONLY OPERATING-POINT RULE.

Model	None	Independent iso	Shared Platt
Llama-4-Scout	67.8/0.076	68.2/0.086	69.5/0.124
LLaVA-7B	75.8/0.001	75.7/0.015	75.8/0.002
Pixtral-12B	54.8/0.224	54.3/0.232	54.0/0.230
Qwen-235B	83.2/0.134	82.1/0.118	81.2/0.118
Qwen2-VL-72B	74.0/0.002	73.3/0.002	74.0/0.002
Qwen2-VL-7B	69.3/0.356	69.2/0.354	69.3/0.472

the utility calculation; the finite-sample result is weaker and more useful: calibration is a model-selection choice, not a guaranteed accuracy improvement. The main empirical failure of uncalibrated confidence is the VLM Log-prob Cascade in Figure 3, not a claim that isotonic wins every aggregate routing metric.

Simple linguistic structure carries much of the signal. Table XI shows that structural features remain close to the full feature set for several models, and question-type-only features improve Qwen-235B to 85.3%, but at 0.298 MB/query versus 0.118 MB/query for the full feature set. TF–IDF helps distinguish templates within a broad class and often improves the cost side of the trade-off, but its gains are corpus-dependent. This supports the motivating hypothesis that language is practically useful while rejecting the stronger claim that surface form fully determines fidelity.

Unseen language is the dominant failure mode. Table IX shows that holding out entire question templates reduces accuracy for every model by 17.4–37.1 points; Qwen-235B falls from 82.1% to 45.0%. Image-grouped folds prevent visual leakage, but they do not by themselves establish robustness to new ways of asking. A deployment needs template-diverse calibration data, shift detection, and a conservative fallback when the question is outside the training language.

The router is small enough to remain outside the remote inference path. The saved Qwen-235B artifact has a 4.12 ms warm median and 5.32 ms p95 on the reference CPU, occupies 234,666 bytes, and peaks at 162 MB process memory. Figure 2 therefore reflects a practical placement: selection can occur where the query and metadata are available, without invoking a visual encoder before acquisition.

D. Failure analysis

The selected representation depends on the target model, not only the wording. Figure 6 shows markedly different fidelity mixtures although every model sees the same question population. Qwen-72B is served mainly by the 32-pixel representation, whereas Pixtral and Qwen-7B require a broader mix of compressed images. Thus the linguistic signal predicts model-specific success; it is not a universal mapping from question type to resolution.

The Qwen2-VL-72B full-resolution drop should be treated as a model/preprocessing interaction, not as evidence that information hurts. Table I shows Qwen2-VL-72B at 73.9% on 32-pixel inputs but 58.8% on full images, a 15.1-point reversal. The error is concentrated in the FloodNet condition-recognition mix where compressed inputs already preserve the coarse scene label. A plausible mechanism is Qwen2-VL's

TABLE XI
LINGUISTIC FEATURE ABLATION AS ACCURACY (%)/MB PER QUERY. STRUCTURAL FEATURES EXCLUDE TF–IDF; TYPE-ONLY RETAINS ONLY BROAD QUESTION FORM.

Model	TF–IDF+struct	Struct only	Type only
Llama-4-Scout	68.2/0.086	68.4/0.089	69.0/0.790
LLaVA-7B	75.7/0.015	76.1/0.001	74.5/0.233
Pixtral-12B	54.3/0.232	53.8/0.229	49.5/0.289
Qwen-235B	82.1/0.118	81.1/0.110	85.3/0.298
Qwen2-VL-72B	73.3/0.002	73.4/0.003	73.7/0.002
Qwen2-VL-7B	69.2/0.354	68.7/0.302	70.7/0.761

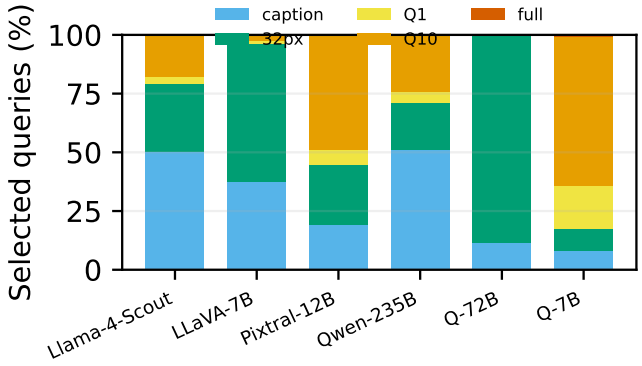


Fig. 6. Test-time VOILA selections by target VLM. Language is shared, but the learned fidelity mixture changes with the VLM's measured behavior.

dynamic image preprocessing on large aerial originals: full-resolution files change tiling, token allocation, and prompt packing relative to the tiny PNG, and may expose texture that is irrelevant to the answer-matching label. We therefore report the anomaly explicitly and avoid using it as a causal claim; resolving it requires model-internal preprocessing traces or a manual answer audit.

Under-acquisition is the consequential reducible error. Table XII distinguishes cheap mistakes from expensive successes. An under-acquired query is wrong even though at least one cached fidelity could answer it; better probability estimation could recover that accuracy. An over-acquired query is correct but spends more bytes than the hindsight oracle; better cost ranking could recover efficiency. The remaining "None" cases cannot be repaired by fidelity routing because the cached VLM fails at every representation.

The oracle gap is not one problem. Table XII shows that exact oracle selection is a minority outcome for several models, yet aggregate accuracy can remain high because different fidelities often answer the same query correctly. Closing the gap therefore requires two distinct interventions: richer language or weak visual metadata to reduce under-acquisition, and sharper calibrated cost differences to reduce over-acquisition. A more complex router that addresses only one may move the wrong axis of Figure 3.

The linguistic signal is not unique to flood questions.

TABLE XII
FAILURE DECOMPOSITION (%). ORACLE IS THE FRACTION ANSWERABLE BY AT LEAST ONE FIDELITY; EXACT SELECTS ITS CHEAPEST SUCCESSFUL TIER; UNDER SELECTS AN INCORRECT TIER DESPITE AN AVAILABLE ANSWER; OVER ANSWERS CORRECTLY WITH MORE BYTES THAN THE CACHED ORACLE; NONE IS UNANSWERABLE AT ALL FIVE TIERS.

Model	Oracle	Exact	Under	Over	None
Llama-4-Scout	88.6	51.8	20.3	16.4	11.4
LLaVA-7B	83.7	33.6	8.0	42.1	16.3
Pixtral-12B	79.4	40.9	25.1	13.4	20.6
Qwen-235B	90.2	59.4	8.2	22.7	9.8
Qwen2-VL-72B	86.3	46.2	13.0	27.1	13.7
Qwen2-VL-7B	84.2	7.8	14.9	61.4	15.8

TABLE XIII
SECONDARY EXTERNAL-VALIDITY EVIDENCE FROM A CACHED MULTI-DATASET STUDY. COST IS ITS NORMALIZED ACQUISITION PROXY, NOT MEASURED FLOODNET BYTES, SO IT IS NOT POOLED WITH THE PRIMARY RESULTS.

Dataset	Models	Accuracy	Normalized cost reduction
VQAV2	6	72.8	34.8%
GQA	6	68.1	36.6%
TEXTVQA	6	89.0	55.6%

Table XIII retains a compact VQA-v2, GQA, and TextVQA check from the broader cached study [32]–[34]. It is secondary evidence because its cost model differs from the measured-byte FloodNet protocol. The consistent savings motivate broader evaluation, but do not substitute for same-protocol replication on additional domains.

VII. DISCUSSION

Pre-retrieval selection is an information decision, not a compression codec. A codec maps an image to fewer bits after the image is available. VOILA instead chooses among representations that already exist at a storage or network boundary. The distinction explains why byte metadata is legitimate input while pixels are not: metadata is part of the index used to decide what to fetch. The policy can select a caption, thumbnail, compressed image, or original without decoding all alternatives.

The method is complementary to model routing. After VOILA acquires one representation, RouteLLM-style logic may still choose a remote model, and an early-exit mechanism may still shorten its computation. Conversely, a model router cannot retroactively remove bytes already retrieved. A joint policy would optimize a pair (f, m) over fidelity and model, but would require calibrated success estimates for every pair and substantially more offline labels.

The 95% rule is a deployment contract rather than a universal objective. We use it to prevent the byte penalty from selecting a cheap but unacceptably inaccurate point. A medical archive may demand a higher floor; a best-effort search interface may accept a lower one. Equation 2 supports either

choice, but the operating point must be selected without test data and reported with the corresponding failure cost.

Fallbacks should respond to linguistic uncertainty. The template-holdout experiment suggests a practical safeguard: route questions far from the calibration language to a conservative fidelity rather than trusting an extrapolated probability. Conformal abstention, nearest-template distance, or a small online calibration buffer could implement this behavior. The current experiments quantify the need for such a mechanism but do not evaluate one.

VIII. Related Work

Visual question answering and linguistic difficulty. VQA established free-form language queries over images [35]; VQA-v2, GQA, and TextVQA expose complementary language priors, compositional reasoning, and OCR requirements [32]–[34]. Analyses consistently find question type predictive of difficulty and failure [20], [21]. VOILA uses that regularity for a different decision: estimating the information needed before acquiring the visual input.

Pre-retrieval prediction and multimodal retrieval. Pre-retrieval query-performance prediction estimates retrieval difficulty from lexical and syntactic features without executing retrieval [22]–[24]. RAG and multimodal RAG retrieve external text or images before generation [36]–[38]. These systems typically choose which item to retrieve at a fixed representation quality. VOILA extends the pre-retrieval view to choosing the fidelity of an already identified visual asset.

Adaptive inference and routing. FrugalGPT, RouteLLM, hybrid routing, and related routers choose among model tiers after the input is available [6]–[8], [39]. Early-exit methods shorten computation when intermediate confidence is sufficient [9], [10], [40]. These methods optimize how a retrieved input is processed. Pre-retrieval fidelity selection instead optimizes what input is acquired, and can precede any of them.

Adaptive visual fidelity. Learned image compression, bandwidth-adaptive video analytics, and edge offloading reduce representation, sampling, or compute after a stream or model stage is available [11]–[16], [41], [42]. Semantic communication similarly transmits task-relevant meaning rather than raw bits, but typically learns an encoder over the observed signal [17]. Recent multimodal work adapts visual token budgets or resolution inside the VLM, as in Matryoshka multimodal models and FastV-style pruning [18], [19]. CARES selects resolution using a compact VLM over the image, and prompt-guided prefiltering identifies relevant regions after image access [43], [44]. VOILA targets the stricter boundary at which pixels are not yet available: it chooses which cached representation should be acquired at all. Tiered mobile and cloud storage further motivates treating acquisition as a first class cost [45]–[47].

Value of information and active acquisition. Value-of-information methods choose whether the expected benefit of evidence justifies its cost [48]. Cost-sensitive feature acquisition and active multimodal acquisition apply related principles to tabular or sequential observations [49], [50]. VOILA specializes this decision to mutually available image representations, using language-conditioned VLM correctness as value and measured payload bytes as cost.

IX. Limitations

Language does not uniquely determine visual need. The template-holdout losses are large, and visually ambiguous questions can require different fidelity despite identical wording. The current policy has no access to scene content before selection and therefore cannot resolve such cases.

Cached representations define the accounting boundary. Captions and VLM answers are reused from prior runs. Caption creation latency and energy are excluded because captions are generated once during ingestion. This assumption is appropriate for repeatedly queried archives, but not for a one-shot stream with no preprocessing. Correctness also inherits the cached answer-matching procedure and service version.

The network experiment is a serialization simulation. CellReplay traces are public 5G proxies rather than disaster deployments. We model recorded delivery opportunities, delay offsets, packetization, and circular start phases, but omit TCP state, protocol headers, retransmission semantics, radio energy, downlink, VLM latency, and human response. The reported DQA is consequently not an end-to-end operational latency claim.

One dataset anchors the measured-byte claim. The primary protocol is FloodNet-centered because it provides the complete common population across six cached VLMs. The secondary datasets use an older normalized cost model. Same-protocol measured-byte replication is needed before generalizing effect sizes to document, medical, or consumer-photo retrieval.

X. Conclusion

The fidelity of visual evidence is an acquisition decision, not merely a VLM preprocessing parameter. VOILA shows that question language can guide this decision before retrieval or transmission: calibrated, fidelity-specific success estimates identify a useful accuracy–byte frontier without accessing pixels or invoking the remote model. On the measured FloodNet population this removes 97.5% of full-image bytes on average and turns the savings into substantially more correct answers by a one-second upload deadline. The fixed-policy and template-shift results also define the boundary of the contribution. Language is a strong prior over information need, not a complete substitute for visual context. More broadly, the work argues that a multimodal system should estimate the value of evidence before paying to acquire it.

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